

## 1.4 Curves in $\mathbf{R}^3$

Let  $I$  be an open interval in the real line  $\mathbf{R}$ . We shall interpret this liberally to include not only the usual finite open interval  $a < t < b$  ( $a, b$  real numbers), but also the infinite types  $a < t$  (a half-line to  $+\infty$ ),  $t < b$  (a half-line to  $-\infty$ ), and also the whole real line.

One can picture a curve in  $\mathbf{R}^3$  as a trip taken by a moving point  $\alpha$ . At each “time”  $t$  in some open interval,  $\alpha$  is located at the point

$$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$$

in  $\mathbf{R}^3$ . In rigorous terms then,  $\alpha$  is a function from  $I$  to  $\mathbf{R}^3$ , and the real-valued functions  $\alpha_1, \alpha_2, \alpha_3$  are its *Euclidean coordinate functions*. Thus we write  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ , meaning, of course, that

$$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t)) \quad \text{for all } t \text{ in } I.$$

We define the function  $\alpha$  to be *differentiable* provided its (real-valued) coordinate functions are differentiable in the usual sense.

**4.1 Definition** A *curve* in  $\mathbf{R}^3$  is a differentiable function  $\alpha: I \rightarrow \mathbf{R}^3$  from an open interval  $I$  into  $\mathbf{R}^3$ .

We shall give several examples of curves, which will be used in Chapter 2 to experiment with results on the geometry of curves.

**4.2 Example** (1) *Straight line*. A line is the simplest type of curve in Euclidean space; its coordinate functions are linear (in the sense  $t \rightarrow at + b$ , not in the homogeneous sense  $t \rightarrow at$ ). Explicitly, the curve  $\alpha: \mathbf{R} \rightarrow \mathbf{R}^3$  such that

$$\alpha(t) = \mathbf{p} + t\mathbf{q} = (p_1 + tq_1, p_2 + tq_2, p_3 + tq_3) \quad (\mathbf{q} \neq 0)$$

is the *straight line* through the point  $\mathbf{p} = \alpha(0)$  in the  $\mathbf{q}$  direction.

(2) *Helix*. (Fig. 1.7). The curve  $t \rightarrow (a \cos t, a \sin t, 0)$  travels around a circle of radius  $a > 0$  in the  $xy$  plane of  $\mathbf{R}^3$ . If we allow this curve to rise (or fall) at a constant rate, we obtain a *helix*  $\alpha: \mathbf{R} \rightarrow \mathbf{R}^3$ , given by the formula

$$\alpha(t) = (a \cos t, a \sin t, bt)$$

where  $a > 0, b \neq 0$ .

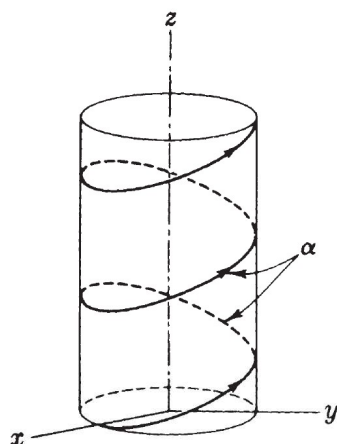


FIG. 1.7

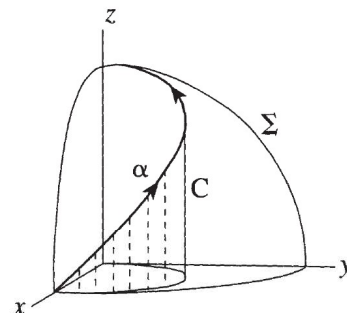


FIG. 1.8

(3) The curve

$$\alpha(t) = \left( 1 + \cos t, \sin t, 2 \sin \frac{t}{2} \right) \text{ for all } t$$

has a noteworthy property: Let  $C$  be the cylinder in  $\mathbf{R}^3$  over the circle in the  $xy$  plane with center at  $(1, 0, 0)$  and radius 1. Then  $\alpha$  perpetually travels the route sliced from  $C$  by the sphere  $\Sigma$  with radius 2 and center at the origin. A segment of this route is shown in Fig. 1.8.

(4) The curve  $\alpha: \mathbf{R} \rightarrow \mathbf{R}^3$  such that

$$\alpha(t) = (e^t, e^{-t}, \sqrt{2}t)$$

shares with the helix in (2) the property of rising constantly. However, it lies over the hyperbola  $xy = 1$  in the  $xy$  plane instead of over a circle.

(5) The 3-curve  $\alpha: \mathbf{R} \rightarrow \mathbf{R}^3$  is defined by

$$\alpha(t) = (3t - t^3, 3t^2, 3t + t^3).$$

If the coordinate functions of a curve are simple enough, its shape in  $\mathbf{R}^3$  can be found, at least approximately, by plotting a few points. We could get a reasonable picture of curve  $\alpha$  for  $0 \leq t \leq 1$  by computing  $\alpha(t)$  for  $t = 0, \frac{1}{10}, \frac{1}{2}, \frac{9}{10}, 1$ .

If we visualize a curve  $\alpha$  in  $\mathbf{R}^3$  as a moving point, then at every time  $t$  there is a tangent vector at the point  $\alpha(t)$  that gives the instantaneous velocity of  $\alpha$  at that time.  $\square$

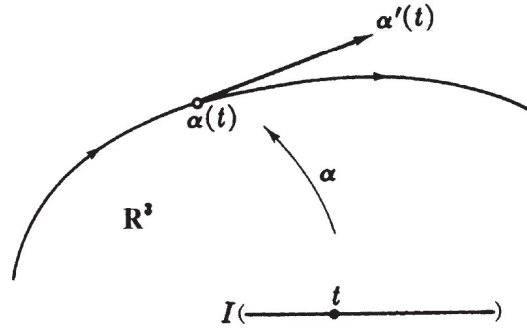


FIG. 1.9

**4.3 Definition** Let  $\alpha: I \rightarrow \mathbf{R}^3$  be a curve in  $\mathbf{R}^3$  with  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ . For each number  $t$  in  $I$ , the *velocity vector* of  $\alpha$  at  $t$  is the tangent vector

$$\alpha'(t) = \left( \frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}$$

at the point  $\alpha(t)$  in  $\mathbf{R}^3$  (Fig. 1.9).

This definition can be interpreted geometrically as follows. The derivative at  $t$  of a real-valued function  $f$  on  $\mathbf{R}$  is given by

$$\frac{df}{dt}(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

This formula still makes sense if  $f$  is replaced by a curve  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ . In fact,

$$\frac{1}{\Delta t}(\alpha(t + \Delta t) - \alpha(t)) = \left( \frac{\alpha_1(t + \Delta t) - \alpha_1(t)}{\Delta t}, \frac{\alpha_2(t + \Delta t) - \alpha_2(t)}{\Delta t}, \frac{\alpha_3(t + \Delta t) - \alpha_3(t)}{\Delta t} \right).$$

This is the vector from  $\alpha(t)$  to  $\alpha(t + \Delta t)$ , scalar multiplied by  $1/\Delta t$  (Fig. 1.10).

Now, as  $\Delta t$  gets smaller,  $\alpha(t + \Delta t)$  approaches  $\alpha(t)$ , and in the limit as  $\Delta t \rightarrow 0$ , we get a vector *tangent* to the curve  $\alpha$  at the point  $\alpha(t)$ , namely,

$$\left( \frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right).$$

As the figure suggests, the point of application of this vector must be the point  $\alpha(t)$ . Thus the standard limit operation for derivatives gives rise to our definition of the velocity of a curve.

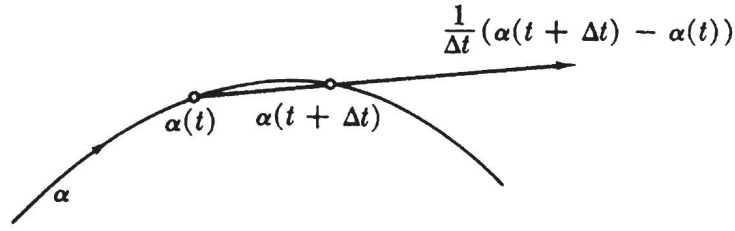


FIG. 1.10

An application of the identity

$$(v_1, v_2, v_3)_p = \sum v_i U_i(\mathbf{p})$$

to the velocity vector  $\alpha'(t)$  at  $t$  yields the alternative formula

$$\alpha'(t) = \sum \frac{d\alpha_i}{dt}(t) U_i(\alpha(t)).$$

For example, the velocity of the straight line  $\alpha(t) = \mathbf{p} + t\mathbf{q}$  is

$$\alpha'(t) = (q_1, q_2, q_3)_{\alpha(t)} = \mathbf{q}_{\alpha(t)}.$$

The fact that  $\alpha$  is straight is reflected in the fact that all its velocity vectors are parallel; only the point of application changes as  $t$  changes.

For the helix

$$\alpha(t) = (a \cos t, a \sin t, bt),$$

the velocity is

$$\alpha'(t) = (-a \sin t, a \cos t, b)_{\alpha(t)}.$$

The fact that the helix rises constantly is shown by the constancy of the  $z$  coordinate of  $\alpha'(t)$ .

Given any curve, it is easy to construct new curves that follow the same route.

**4.4 Definition** Let  $\alpha: I \rightarrow \mathbf{R}^3$  be a curve. If  $h: J \rightarrow I$  is a differentiable function on an open interval  $J$ , then the composite function

$$\beta = \alpha(h): J \rightarrow \mathbf{R}^3$$

is a curve called a *reparametrization* of  $\alpha$  by  $h$ .

For each  $s \in J$ , the new curve  $\beta$  is at the point  $\beta(s) = \alpha(h(s))$  reached by  $\alpha$  at  $h(s)$  in  $I$  (Fig. 1.11). Thus  $\beta$  represents a different trip over at least part of the route of  $\alpha$ .

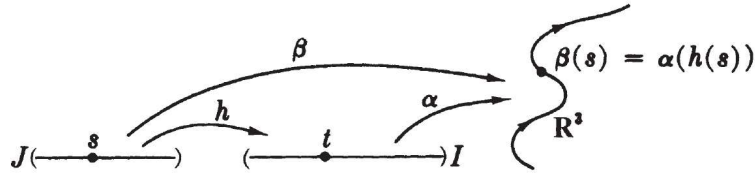


FIG. 1.11

To compute the coordinates of  $\beta$ , simply substitute  $t = h(s)$  into the coordinates  $\alpha_1(t)$ ,  $\alpha_2(t)$ ,  $\alpha_3(t)$  of  $\alpha$ . For example, suppose

$$\alpha(t) = (\sqrt{t}, t\sqrt{t}, 1-t) \text{ on } I: 0 < t < 4.$$

If  $h(s) = s^2$  on  $J: 0 < s < 2$ , then the reparametrized curve is

$$\beta(s) = \alpha(h(s)) = \alpha(s^2) = (s, s^3, 1-s^2).$$

The following lemma relates the velocities of a curve and of a reparametrization.

**4.5 Lemma** If  $\beta$  is the reparametrization of  $\alpha$  by  $h$ , then

$$\beta'(s) = (dh/ds)(s)\alpha'(h(s)).$$

**Proof.** If  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ , then

$$\beta(s) = \alpha(h(s)) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s))).$$

Using the “prime” notation for derivatives, the chain rule for a composition of real-valued functions  $f$  and  $g$  reads  $(g(f))' = g'(f) \cdot f'$ . Thus, in the case at hand,

$$\alpha_i(h)'(s) = \alpha_i'(h(s)) \cdot h'(s).$$

By the definition of velocity, this yields

$$\begin{aligned} \beta'(s) &= \alpha(h)'(s) \\ &= (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s)) \\ &= h'(s)\alpha'(h(s)). \end{aligned}$$

◆

According to this lemma, to obtain the velocity of a reparametrization of  $\alpha$  by  $h$ , first reparametrize  $\alpha'$  by  $h$ , then scalar multiply by the derivative of  $h$ .

Since velocities are tangent vectors, we can take the derivative of a function with respect to a velocity.



**4.6 Lemma** Let  $\alpha$  be a curve in  $\mathbf{R}^3$  and let  $f$  be a differentiable function on  $\mathbf{R}^3$ . Then

$$\alpha'(t)[f] = \frac{d(f(\alpha))}{dt}(t).$$

**Proof.** Since

$$\alpha' = \left( \frac{d\alpha_1}{dt}, \frac{d\alpha_2}{dt}, \frac{d\alpha_3}{dt} \right)_\alpha,$$

we conclude from Lemma 3.2 that

$$\alpha'(t)[f] = \sum \frac{\partial f}{\partial x_i}(\alpha(t)) \frac{d\alpha_i}{dt}(t).$$

But the composite function  $f(\alpha)$  may be written  $f(\alpha_1, \alpha_2, \alpha_3)$ , and the chain rule then gives exactly the same result for the derivative of  $f(\alpha)$ . ♦

By definition,  $\alpha'(t)[f]$  is the rate of change of  $f$  along the line through  $\alpha(t)$  in the  $\alpha'(t)$  direction. (If  $\alpha'(t) \neq 0$ , this is the tangent line to  $\alpha$  at  $\alpha(t)$ ; see Exercise 9.) The lemma shows that this rate of change is the same as that of  $f$  along the curve  $\alpha$  itself.

Since a curve  $\alpha: I \rightarrow \mathbf{R}^3$  is a function, it makes sense to say that  $\alpha$  is one-to-one; that is,  $\alpha(t) = \alpha(t_1)$  only if  $t = t_1$ . Another special property of curves is periodicity: A curve  $\alpha: \mathbf{R} \rightarrow \mathbf{R}^3$  is *periodic* if there is a number  $p > 0$  such that  $\alpha(t + p) = \alpha(t)$  for all  $t$ —and the smallest such number  $p$  is then called the *period* of  $\alpha$ .

From the viewpoint of calculus, the most important condition on a curve  $\alpha$  is that it be *regular*, that is, have all velocity vectors different from zero. Such a curve can have no corners or cusps.

The following remarks about curves (offered without proof) describe another familiar way to formulate the concept of “curve.” If  $f$  is a differentiable real-valued function on  $\mathbf{R}^2$ , let

$$C: f = a$$

be the set of all points  $\mathbf{p}$  in  $\mathbf{R}^2$  such that  $f(\mathbf{p}) = a$ . Now, if the partial derivatives  $\partial f / \partial x$  and  $\partial f / \partial y$  are never simultaneously zero at any point of  $C$ , then  $C$  consists of one or more separate “components,” which we shall call *Curves*.† For example,  $C: x^2 + y^2 = r^2$  is the circle of radius  $r$  centered at the

† The capital C distinguishes this notion from a (parametrized) curve  $\alpha: I \rightarrow \mathbf{R}^2$ .

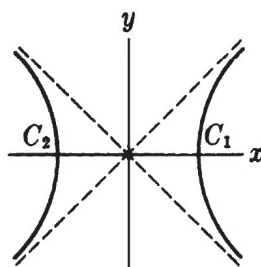


FIG. 1.12

origin of  $\mathbf{R}^2$ , and the hyperbola  $C: x^2 - y^2 = r^2$  splits into two Curves (“branches”)  $C_1$  and  $C_2$  as shown in Fig. 1.12.

Every Curve  $C$  is the route of many regular curves, called *parametrizations* of  $C$ . For example, the curve

$$\alpha(t) = (r \cos t, r \sin t)$$

is a well-known periodic parametrization of the circle given above, and for  $r > 0$  the one-to-one curve

$$\beta(t) = (r \cosh t, r \sinh t)$$

parametrizes the branch  $x > 0$  of the hyperbola.

## Exercises

1. Compute the velocity vector of the curve in Example 4.2(3) for arbitrary  $t$  and for  $t = 0$ ,  $t = \pi/2$ ,  $t = \pi$ , visualizing those on Fig. 1.8.
2. Find the unique curve such that  $\alpha(0) = (1, 0, 5)$  and  $\alpha'(t) = (t^2, t, e^t)$ .
3. Find the coordinate functions of the curve  $\beta = \alpha(h)$ , where  $\alpha$  is the curve in Example 4.2(3) and  $h(s) = \cos^{-1}(s)$  on  $J: 0 < s < 1$ .
4. Reparametrize the curve  $\alpha$  in Example 4.2(4) using  $h(s) = \log s$  on  $J: s > 0$ . Check the equation in Lemma 4.5 in this case by calculating each side separately.
5. Find the equation of the straight line through the points  $(1, -3, -1)$  and  $(6, 2, 1)$ . Does this line meet the line through the points  $(-1, 1, 0)$  and  $(-5, -1, -1)$ ?
6. Deduce from Lemma 4.6 that in the definition of directional derivative (Def. 3.1) the straight line  $t \rightarrow \mathbf{p} + t\mathbf{v}$  can be replaced by any curve  $\alpha$  with *initial velocity*  $\mathbf{v}_p$ , that is, such that  $\alpha(0) = \mathbf{p}$  and  $\alpha'(0) = \mathbf{v}_p$ .